

Assignment - 03

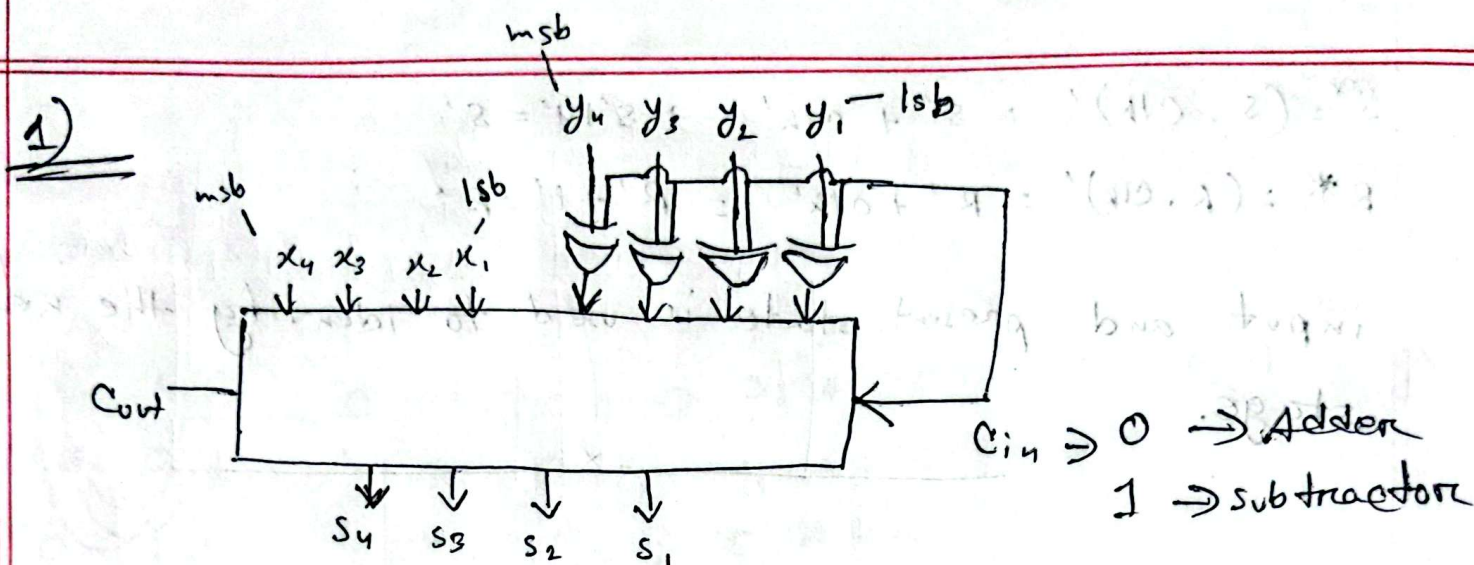
CSE260

Name: Abrar Fahim

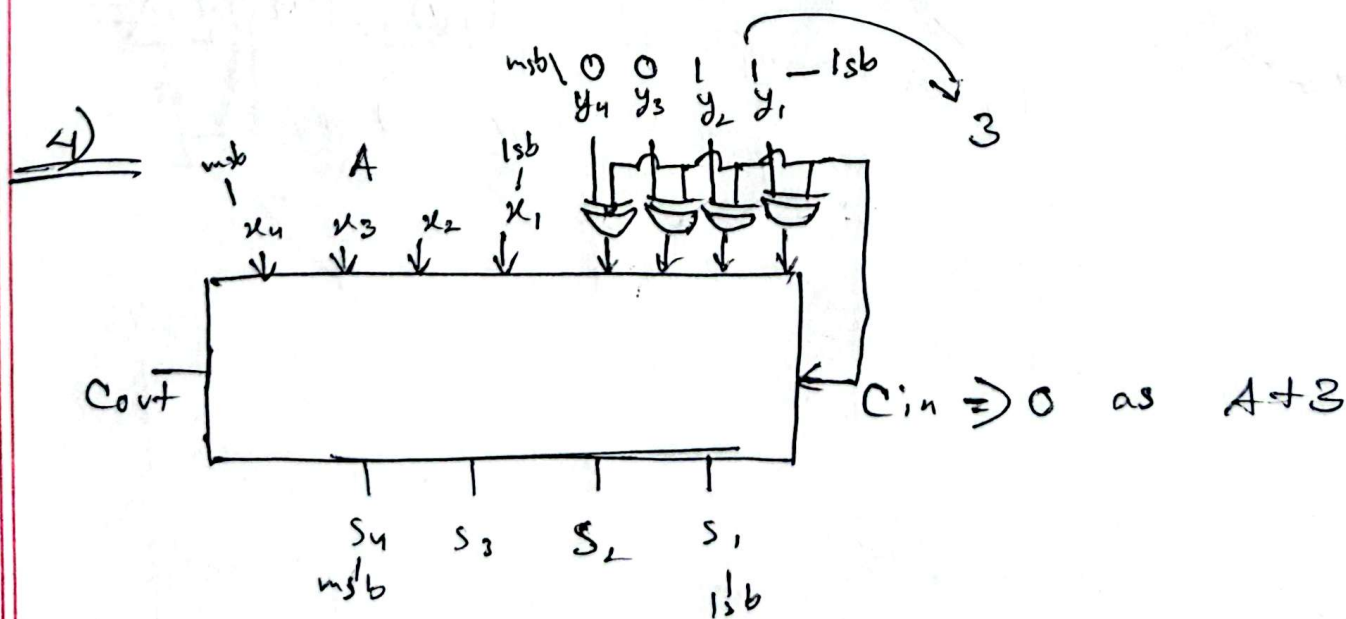
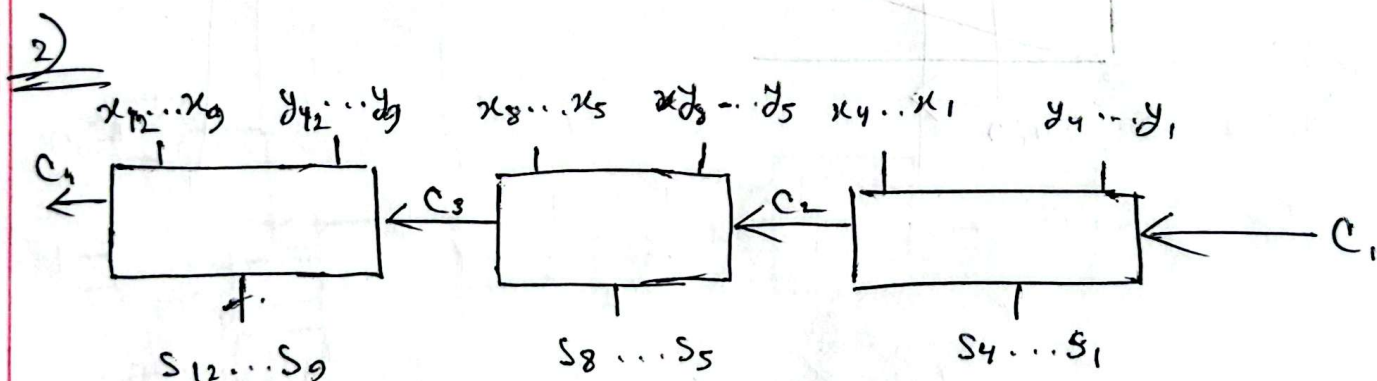
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sec: 02

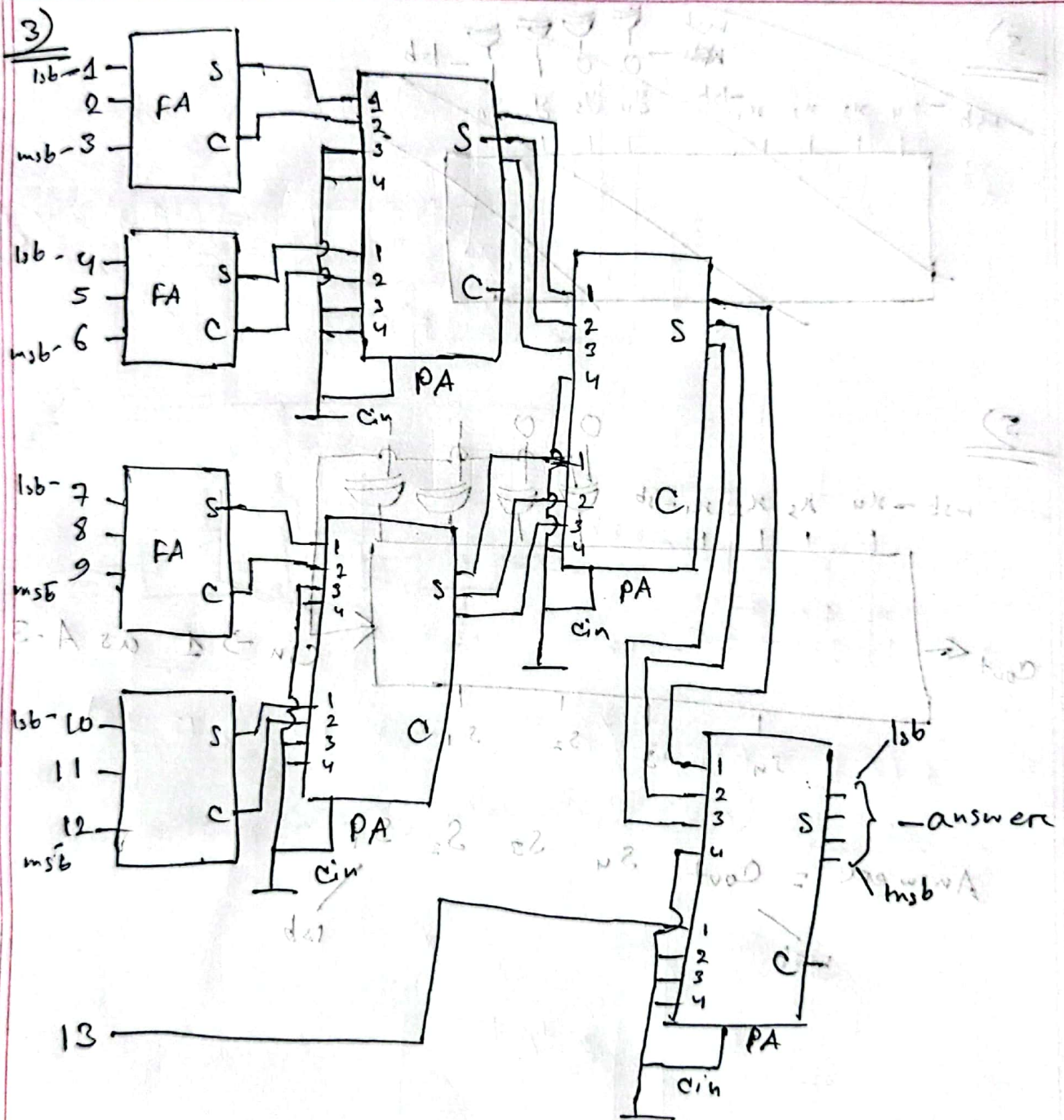
Date : 20/4/25



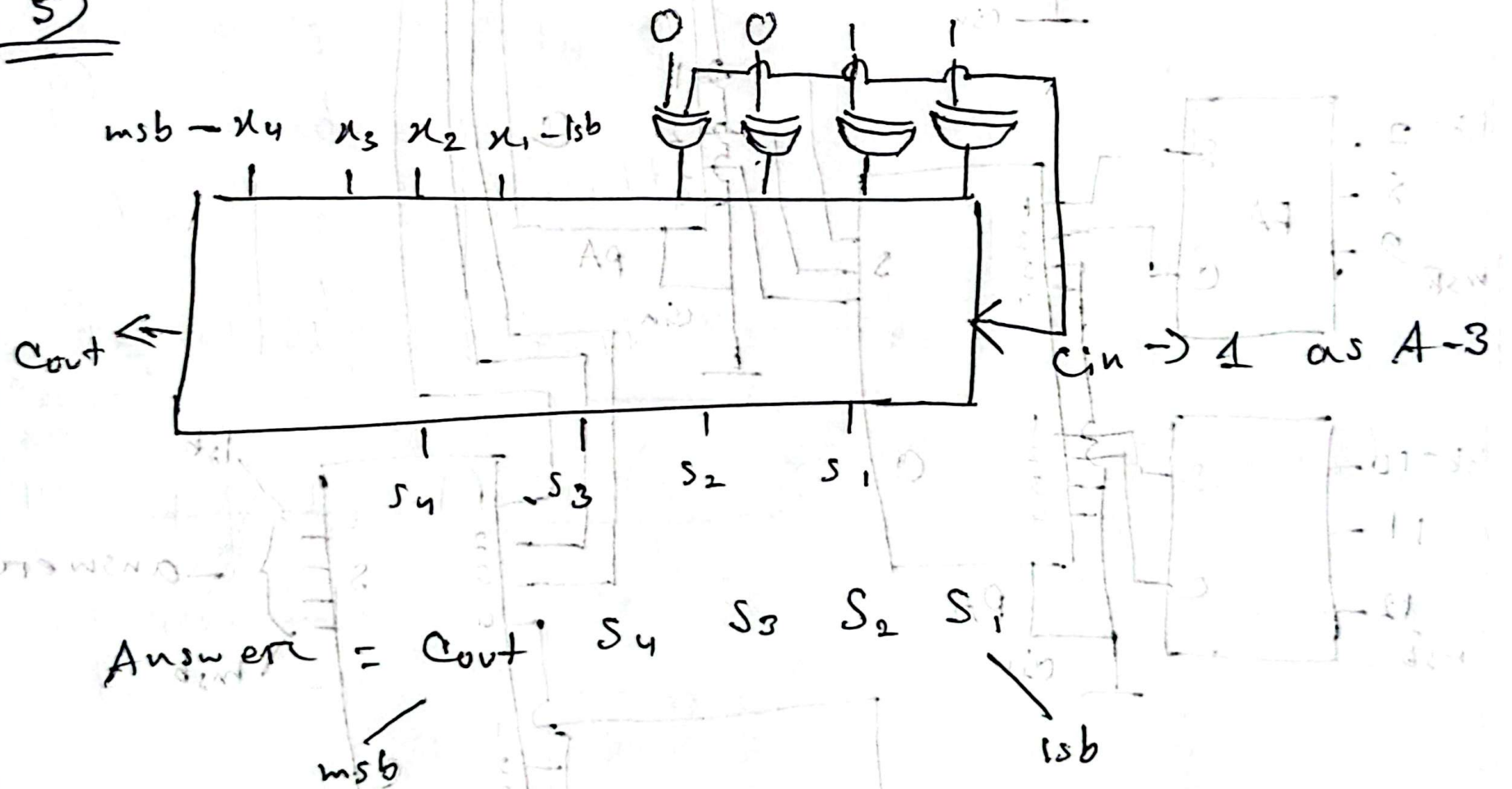
answer $\Rightarrow$ Cout $S_4 S_3 S_2 S_1$
 MSB Lsb



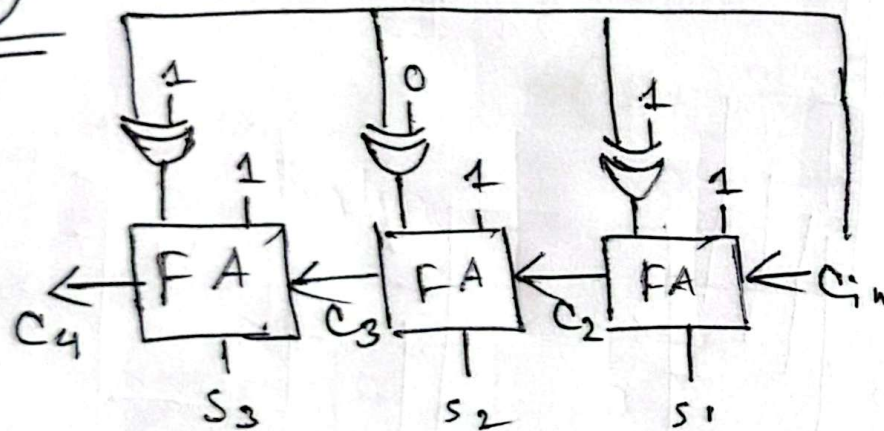
Answer : Cout $S_4 S_3 S_2 S_1$
 msb lsb



5)



G)



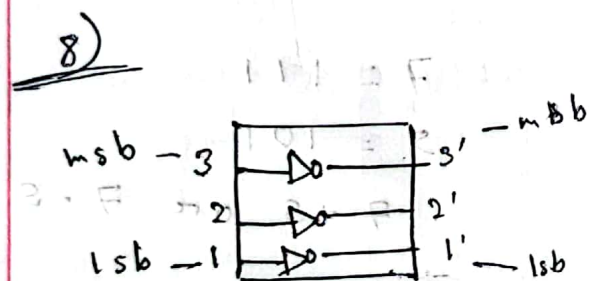
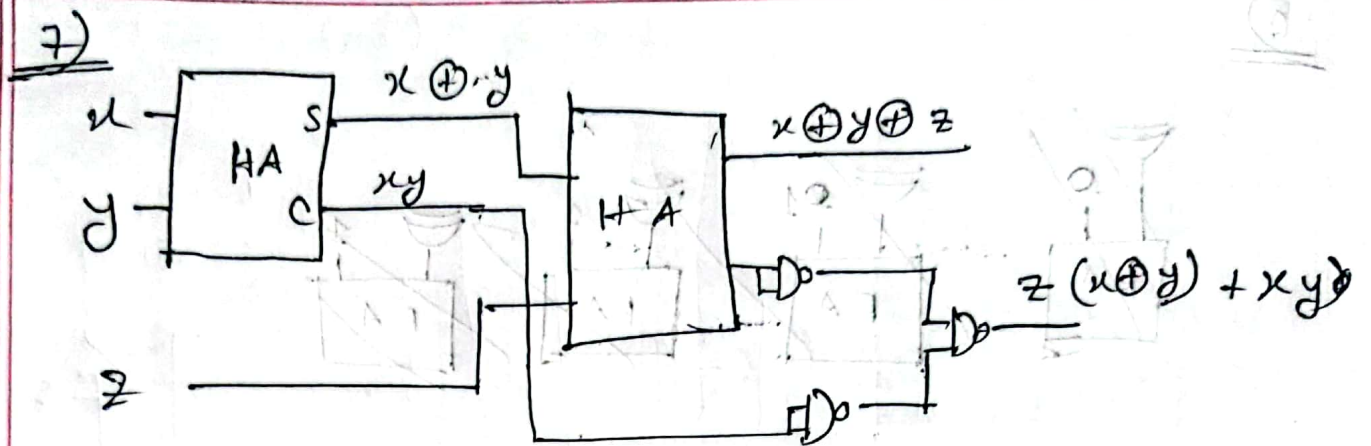
$$7 = 111$$

$$5 = 101$$

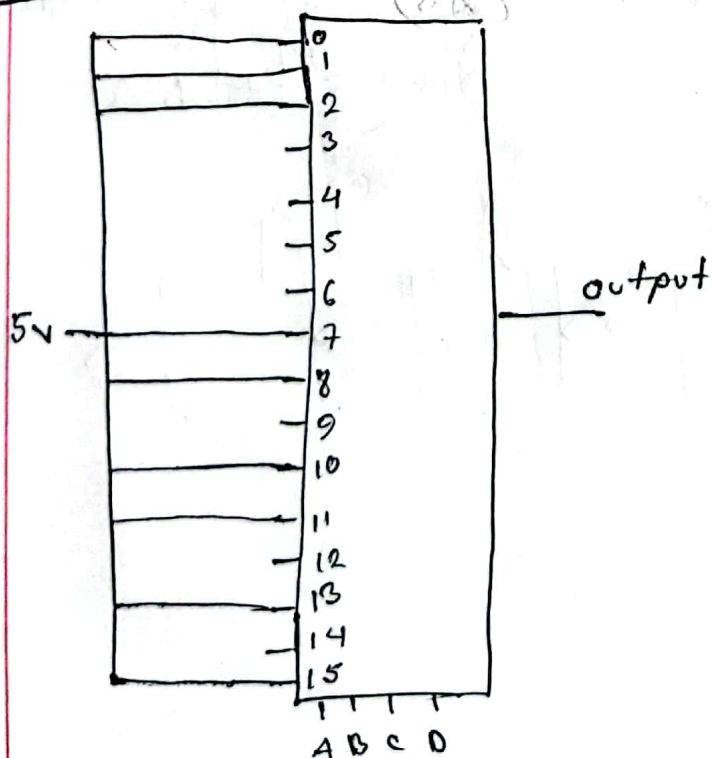
$$7 + 5 \text{ or } 7 - 5$$

if $C_{in} = 1$, then $7 - 5$

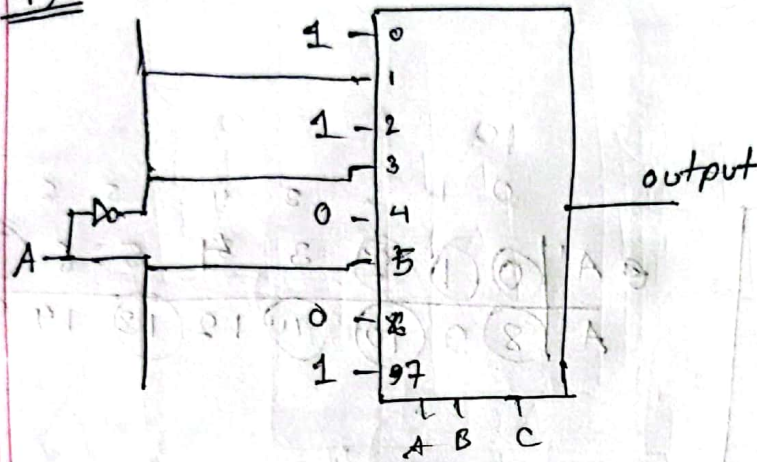
else, $C_{in} = 0$, then $7 + 5$ (Ans)



10) $F(A, B, C, D) = \sum (0, 1, 2, 7, 8, 10, 11, 13, 15)$

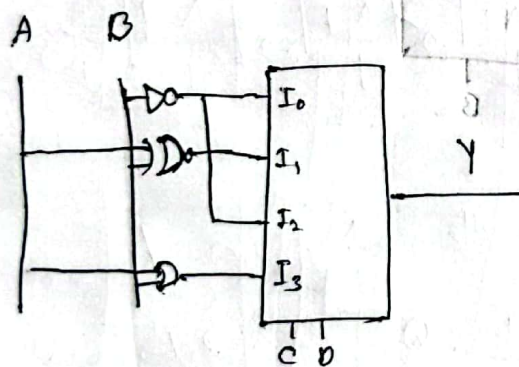


11) $f(A, B, C, D) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15)$



	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15

12) $F(A, B, C, D) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15)$



$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15

$$I_0 = A'B' + AB'$$

$$= B'(A + A') = B'$$

$$I_1 = A'B' + AB$$

$$= A \odot B$$

$$I_2 = A'B' + AB'$$

$$= B'$$

$$I_3 = A'B + AB' + AB$$

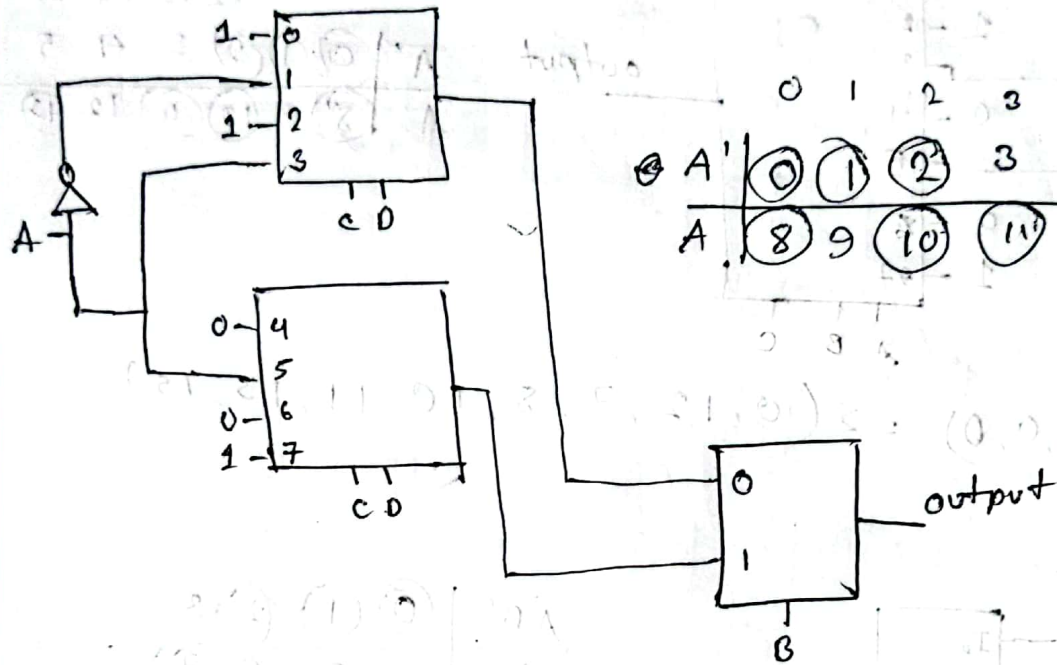
$$= A'B + A(B' + B)$$

$$= A'B + A$$

$$= (A' + A)(B' + A)$$

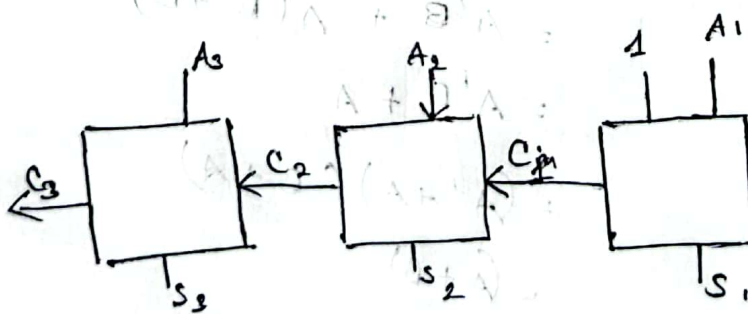
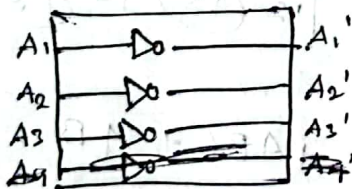
$$= (A + B)$$

13) $F(A, B, C, D) = \Sigma(0, 1, 2, 7, 8, 10, 11, 13, 15)$



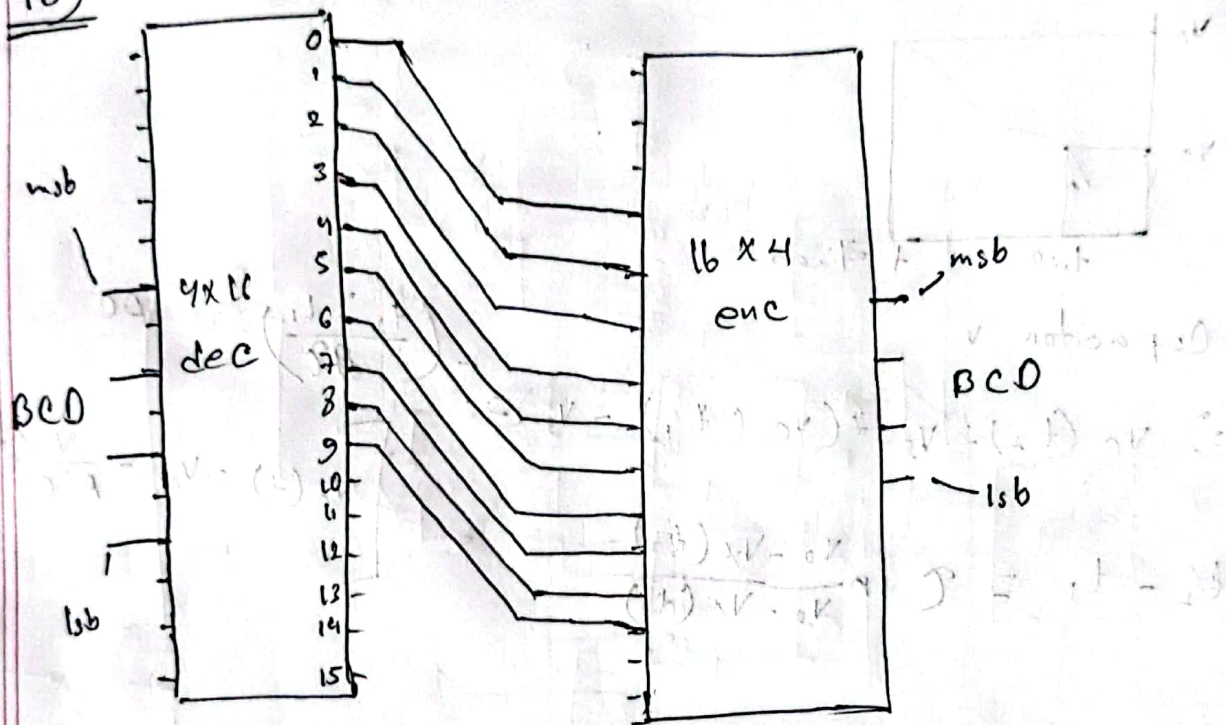
	0	1	2	3	4	5	6	7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15

9)

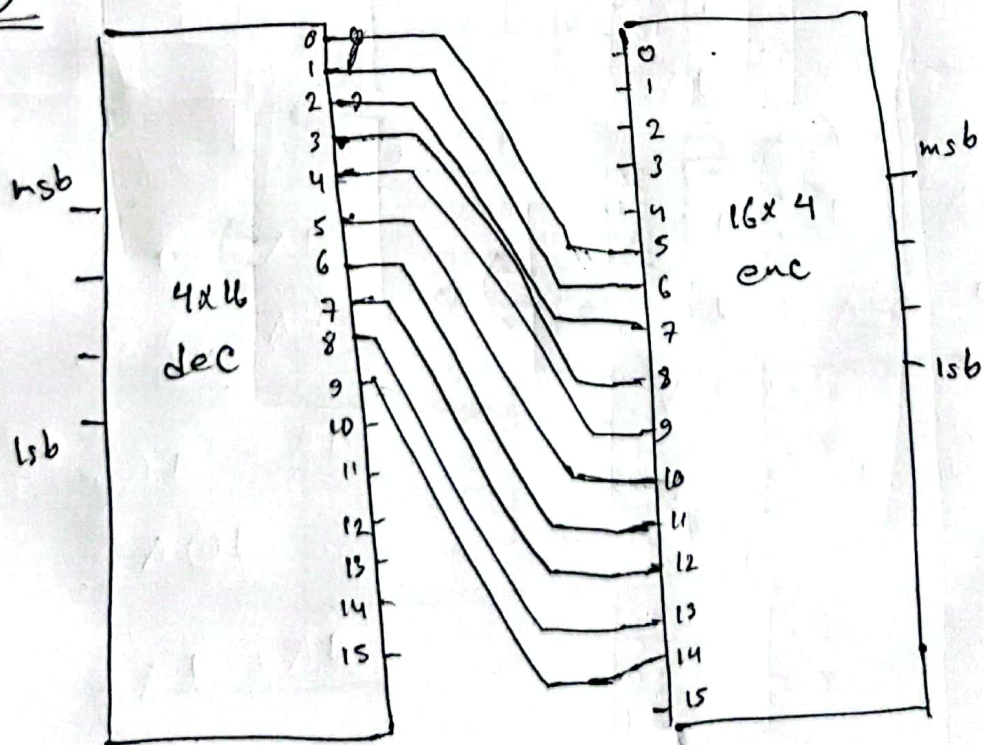


Answer: $C_3 S_3 S_2 S_1$ — lsb

15)

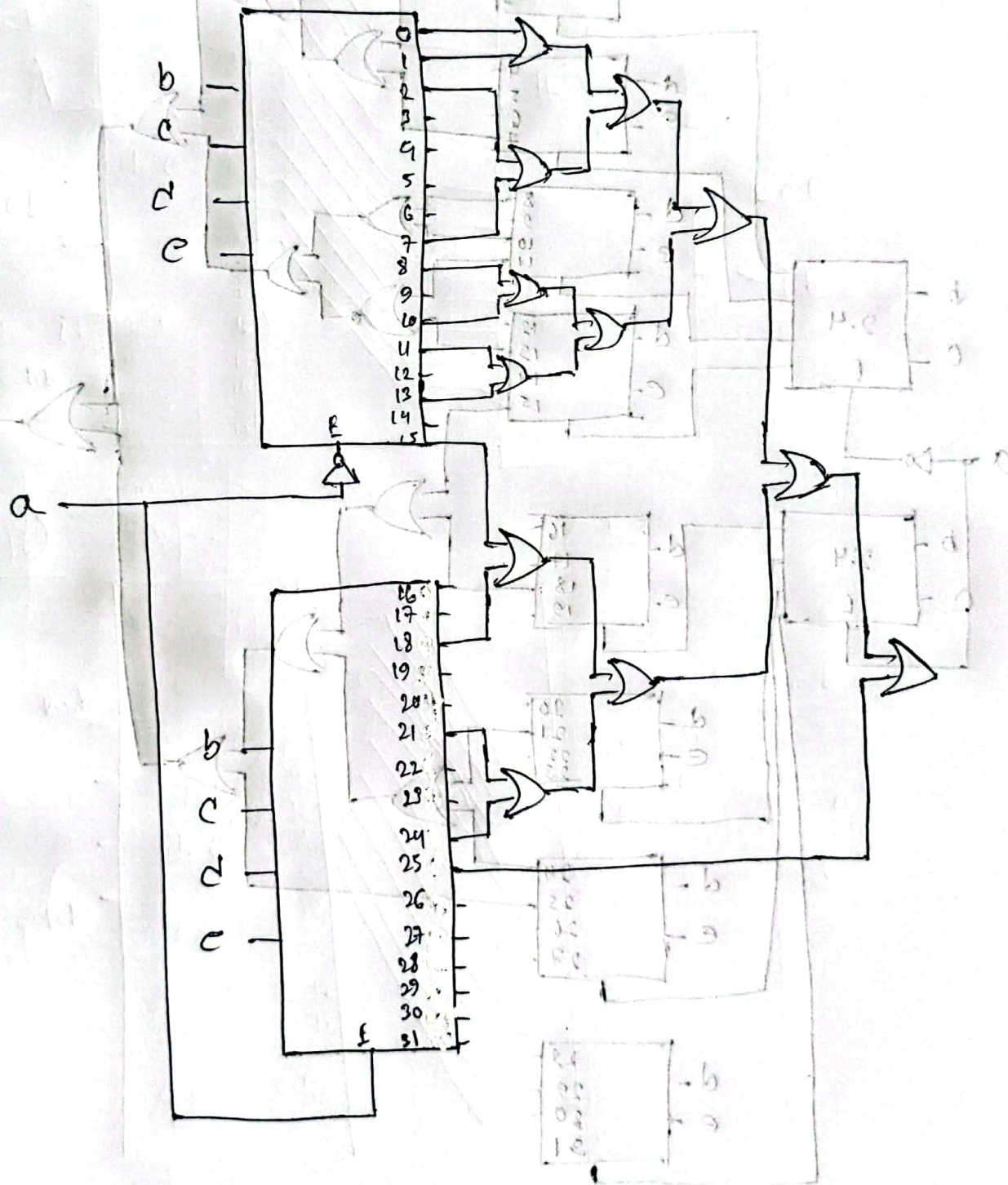


16)



14) $F(A, B, C, D, E) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15, 18, 21, 24, 25)$

a) 4x16



b)

